Aggregate Production Planning Optimization

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1. Problem Definition and System Assumptions

This study investigates the aggregate production planning problem for a single product manufacturing system. The analyzed system represents a hypothetical furniture manufacturing facility producing standard wooden chairs. The primary objective is to minimize total operational costs by determining the optimal production volumes, inventory levels, and workforce adjustments required to satisfy customer demand fully and on time throughout a six month planning horizon. The planning process is executed from a medium term perspective, and production activities are categorized into two distinct modes being regular time production and overtime production.

Demand values were estimated utilizing historical sales data from the planning period. Under the assumption that a trend component exists within the time series, the Holt linear trend method was applied, and the resulting forecasts were utilized as direct inputs for the planning model. These forecasted demand values are treated as deterministic, meaning they are assumed to remain constant throughout the planning horizon. Consequently, the model is structured as a deterministic linear aggregate production planning model devoid of uncertainty.

The model considers a single product and assumes that the quality characteristics of the product remain invariant across all periods. The timely fulfillment of demand is strictly enforced; backordering and lost sales are entirely prohibited. The system allows for product storage, and maintaining this inventory incurs a specific holding cost. The workforce level can be dynamically adjusted between periods through explicit hiring and firing decisions. It is assumed that the production efficiency of an individual employee remains constant throughout the planning horizon and that total production capacity is directly proportional to the active headcount. The mathematical model is strictly linear, and all strategic decisions are made under perfect information.

The planning horizon is established as six periods. The estimated demand values derived from the Holt linear trend method are 800, 900, 1100, 1000, 1200, and 950 units for the respective sequential periods. The maximum regular time production capacity for a single employee in one period is set at 100 units, while the maximum additional production capacity via overtime is limited to 20 units per employee. The unit cost for regular production is 50, the unit cost for overtime production is 70, and the inventory holding cost per unit is 5. The financial penalty for hiring a single employee is established at 1200, whereas the severance cost for firing an employee is 1800. The system initiates with a starting inventory of 200 units and an initial workforce of 10 employees. Furthermore, a strict operational constraint mandates a minimum ending inventory of 150 units at the conclusion of the six month planning horizon.

2. Decision Variables and Parameters

The strategic decisions to be made over the six month planning horizon are expressed through time indexed decision variables and specific system parameters.

- t represents the time index.
- X_t represents the volume of products manufactured during regular working hours in period t.
- Y_t represents the volume of products manufactured utilizing overtime in period t.
- Z_t represents the ending inventory level retained at the conclusion of period t.
- D_t represents the forecasted demand quantity for period t.

- a represents the maximum production capacity a single employee can yield during regular hours within one period.
- b represents the maximum additional production capacity a single employee can yield via overtime within one period.
- A_t represents the total active workforce headcount in period t.
- B_t represents the number of new employees hired during period t.
- C_t represents the number of employees fired during period t.
- c_X represents the unit cost of regular production.
- c_Y represents the unit cost of overtime production.
- c_Z represents the unit holding cost for inventory.
- c_B represents the financial cost incurred per hired employee.
- c_C represents the financial cost incurred per fired employee.

3. Mathematical Model Formulation

The fundamental objective of the model is to minimize the total operational cost accumulated over the six month planning horizon. The total cost consists of regular production costs, overtime production costs, inventory holding costs, hiring costs, and firing costs. Accordingly, the objective function is mathematically defined as the summation of these cost components across every period.

Minimize Total Cost:

$$Min\ M = \sum_{t=1}^{6} c_X X_t + c_Y Y_t + c_Z Z_t + c_B B_t + c_C C_t$$

Inventory Balance Constraint:

$$Z_t = Z_{t-1} + X_t + Y_t - D_t$$

Regular Capacity Constraint:

$$X_t \leq aA_t$$

Overtime Capacity Constraint:

$$Y_t \leq bA_t$$

Workforce Balance Constraint:

$$A_t = A_{t-1} + B_t - C_t$$

Terminal Inventory Constraint:

$$Z_6 \geq 150$$

Domain and Non-Negativity Constraints:

$$X_t, Y_t, Z_t, D_t, a, b, c_X, c_Y, c_Z, c_B, c_C, A_t, B_t, C_t \geq 0$$

$$A_t, B_t, C_t \in \mathbb{Z}$$

$$t = 1,2,3,4,5,6$$

The constraints incorporated into the model ensure that all production, inventory, and workforce decisions remain consistent and operationally viable throughout the planning horizon. The

inventory balance constraint guarantees that demand is fulfilled strictly through current production and available stock. The regular and overtime capacity constraints restrict manufacturing decisions based on the active workforce limitations. The workforce balance constraint ensures that the total headcount fluctuates consistently between periods through explicit hiring and firing variables. The terminal inventory constraint aims to preserve a minimum stock level to support operational continuity immediately following the planning horizon. The non-negativity and integer constraints ensure that all decision variables align seamlessly with physical reality and operational logic.

4. General Strategic Evaluation

In this study, an aggregate production planning model for a six month horizon was formulated using deterministic demand data forecasted via the Holt linear trend method, and it was subsequently solved utilizing a mixed integer linear programming approach. The model successfully generated a feasible solution under all constraints, and the solver verified this result as globally optimal. The absolute minimum value of the objective function was calculated as 309450, and the complete absence of absolute and relative optimality gaps mathematically confirms the strict optimality of this solution.

An analysis of the solution outputs reveals that manufacturing is exclusively executed under regular working hours across all periods, with zero reliance on overtime. This operational behavior is directly explained by the model dismissing overtime as economically unviable due to its significantly higher unit cost compared to regular production. As highlighted in production planning literature, high cost capacity flexibility tools are only selected under absolute necessity; otherwise, optimization models naturally gravitate towards lower cost manufacturing alternatives. The generated results are entirely consistent with this theoretical behavior.

Inventory levels are utilized in a highly restricted manner. The system avoids holding any excess stock during the initial months of the planning period and only accumulates inventory in later periods to strategically balance impending demand surges. The final inventory level perfectly equals the mandated minimum stock constraint, proving that this terminal constraint acts as a binding boundary on the solution. This finding demonstrates that the inventory carrying cost functions highly effectively within the model, actively preventing unwarranted stock accumulation.

Evaluated from a workforce planning perspective, the initial workforce level of 10 personnel is augmented by exactly one hire during the second period, raising the headcount to 11. This specific staffing level is then rigidly maintained throughout the remainder of the planning horizon. Zero layoffs are executed during any period. This specific outcome indicates that the model heavily avoids workforce fluctuations due to the severe financial penalties associated with hiring and firing, favoring a highly stable workforce structure instead. Within aggregate production planning literature, this approach is formally defined as a mixed strategy that simultaneously addresses pure cost minimization and operational stability.

Overall, the obtained solution delivers a balanced production plan that relies heavily on regular manufacturing, incorporates minimal inventory buffering, and aggressively preserves workforce stability. This structural approach aligns perfectly with both the foundational model assumptions and the theoretical expectations of aggregate production planning, thereby providing a highly practical and cost effective strategic roadmap for the enterprise.